LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

Annual Exam Practice Paper 1

Chapter / Topic: Force and Pressure · Microorganisms · Metals and Non-metals · Combustion and Flame · Sound

Name: _____ Roll No.: _____ Date: _____

Time: 2 hours **Max. Marks: 50**

General instructions:

1. All questions are compulsory.
2. Section A: MCQs, 1 mark each. Section B: fill in the blanks and true/false, 1 mark each. Section C: match the columns, 4 marks. Section D: 2 marks each. Section E: 3 marks each. Section F: 4 marks each.
3. Write units in numericals and draw neat, labelled diagrams.
4. This paper is topic-based. Check the topic list against your school's annual exam syllabus before using it.

SECTION A

6 marks

Choose the correct option.

Q1. A rubber sucker stays stuck to a smooth wall because of: **[1]**

(a) friction (b) magnetic force
(c) atmospheric pressure (d) gravity

Q2. The same person exerts the greatest pressure on the ground when: **[1]**

(a) standing on both feet (b) standing on one foot
(c) lying flat on the ground (d) standing on wide skis

Q3. Malaria is caused by a: **[1]**

(a) protozoan (b) bacterium
(c) virus (d) fungus

Q4. The oxide of which element turns moist red litmus paper blue? **[1]**

(a) sulphur (b) carbon
(c) phosphorus (d) magnesium

Q5. Goldsmiths melt gold using the part of the flame that is: **[1]**

(a) the innermost dark zone (b) the middle yellow zone
(c) the wick (d) the outermost blue zone

Q6. A sound is made shriller (higher in pitch) by increasing its: **[1]**

(a) amplitude (b) frequency
(c) loudness (d) time period

SECTION B

8 marks

Q7–Q10: fill in the blanks. Q11–Q14: write True or False.

Q7. The force that opposes motion between two surfaces in contact is called _____. **[1]**

Q8. The process of heating milk to about 70 °C and then chilling it quickly is called _____. **[1]**

Q9. Metals that can be drawn into thin wires are said to be _____. **[1]**

Q10. The lowest temperature at which a substance catches fire is called its _____. **[1]**

Q11. Liquids exert pressure only on the bottom of their container. **[1]**

Q12. Viruses can reproduce only inside the cells of a host organism. **[1]**

Q13. Carbon dioxide can be used to put out a fire in electrical equipment. **[1]**

Q14. Sound travels faster in air than through a solid such as iron. **[1]**

SECTION C

4 marks

Q15. Match each microorganism in Column A with its use in Column B.
Column A: (i) Rhizobium (ii) Yeast (iii) Lactobacillus (iv) Penicillium (a mould)
Column B: (a) setting curd (b) making an antibiotic (c) fixing nitrogen in the soil (d) making bread rise **[4]**

SECTION D

12 marks

Q16. A brick weighs 30 N and measures 0.2 m × 0.1 m × 0.05 m. Find the pressure it exerts on the ground when it rests on (a) its largest face and (b) its smallest face. **[2]**

Q17. Give two examples each of contact forces and non-contact forces from daily life. **[2]**

Q18. What is a vaccine? Who first developed a vaccine, and for which disease? **[2]**

Q19. Why does a heap of green leaves not catch fire easily, while dry leaves catch fire quickly? **[2]**

Q20. Give two uses of metals and two uses of non-metals in daily life. **[2]**

Q21. What is noise pollution? Give two of its harmful effects on health. **[2]**

SECTION E

12 marks

Q22. Describe a simple activity to show that a liquid presses on the walls of its container, and that this pressure increases with depth. **[3]**

Q23. For each disease, name the type of microorganism that causes it and how it spreads: (a) cholera, (b) dengue, (c) tuberculosis. **[3]**

Q24. Draw a neat, labelled diagram of a candle flame showing its three zones. Write which zone is the hottest and which is the least hot. **[3]**

Q25. Element X is a shiny solid that conducts electricity. It burns in air to form an oxide. The oxide dissolves in water to give a solution that turns red litmus blue.
(a) Is X a metal or a non-metal?
(b) Is its oxide acidic or basic?
(c) Name one element that fits, and write the word equations for the two reactions. **[3]**

SECTION F

8 marks

Q26. Answer the following: **[4]**

(i) A guitar string vibrates 600 times in 2 seconds. Find its frequency and time period. Can a human hear this sound? **[2]**

(ii) A dog whistle seems silent to us, but a dog responds to it. Explain why. What are such sounds called? **[2]**

Q27. Answer the following: **[4]**

(i) What is the calorific value of a fuel? Which has the highest calorific value: LPG, petrol, coal or wood? **[2]**

(ii) How does water put out a fire? Why must water not be used on a fire in electrical equipment? **[2]**

~ All the best for your annual exam! ~